

Total Questions: 30

LEVEL – II

Time: 1.5 hr.
Total Marks: 70

SECTION - 1

- This section contains **Twelve (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 2

- This section contains **Six (06)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each correct answer you will get **TWO** marks, if all the options are correct. If the correct answers chosen are less than the actual correct answers, partial marking (basis on weightage) will be given.
- There are **NO NEGATIVE** marks for wrong answers.

SECTION - 3

- This section contains **NINE (09)** questions of **PASSAGE TYPE OR COMPREHENSION TYPE**.
- Under each paragraph there will be three questions that has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 4

- This section contains **THREE (03)** questions. These questions are **MATCHING TYPE BASED QUESTIONS**.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers

Section - 1

1. A milkman is suspected of adding water to the milk he sells. A student uses a Lactometer to test the milk. How does the Lactometer help in detecting this adulteration?

- A) It changes colour if there is water in the milk.
- B) It measures the temperature of the milk.
- C) It measures the density of the milk; adding water makes the milk less dense.
- D) It counts the number of bacteria present in the milk.

2. A chemist prepares three mixtures:

- Salt dissolved in water
- Sand mixed with sugar
- Oil mixed with water

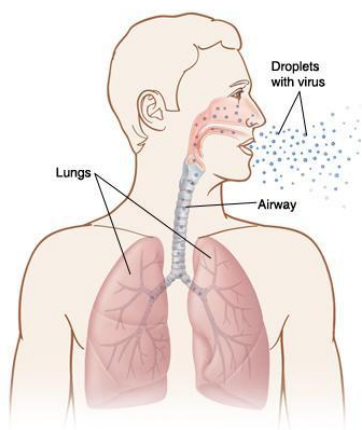
Which of the following statements is correct?

- A) Mixture 1 is heterogeneous because salt is visible in water.
- B) Mixture 2 is homogeneous because both solids mix together.
- C) Mixture 3 is immiscible because the liquids form separate layers.
- D) Mixture 1 is immiscible because it is a liquid solution.

3. A mountain climber carries a 10 kg backpack from the base of a mountain to the peak. Even if the climber takes a very long, winding path, the work done against gravity depends only on what?



- (A) The speed at which the climber moves.
(B) The amount of food the climber ate.
(C) The total distance walked by the climber.
(D) The vertical height of the mountain.
4. The diagram shows disease spread through air by coughing or sneezing. Which disease is MOST likely spread this way?



- (A) Diabetes
(B) Malaria
(C) Rickets
(D) Common cold
5. The diagram shows one plant taking support and another spreading on ground. Which pair correctly identifies them?



- A) Tree and shrub B) Herb and shrub
C) Climber and creeper D) Creeper and tree
6. You are lost in the woods and use a GPS mapping app on your phone to find your way home, while checking a live radar map to avoid a rainstorm. Which combination of artificial satellite jobs are you actively relying on simultaneously?
- A) Weather and Lunar Observation
B) Navigation and Space Exploration
C) Navigation and Weather Monitoring
D) Communication and Earth Observation
7. A mixture contains iron filings, chalk powder, and sugar. Which of the following sequences would be the most effective way to separate all three components?
- A) Filtration → Magnetic Separation → Evaporation
B) Magnetic Separation → Filtration → Evaporation
C) Evaporation → Filtration → Magnetic Separation
D) Handpicking → Evaporation → Filtration
8. "Based on the physical properties of a rock sample—specifically the presence of visible particles, distinct layering, and embedded fossils—to which of the three main rock categories does it belong, and what does its structure indicate about its formation?"



- A) Granite B) Shale
C) Marble D) Obsidian

9. A snake and an earthworm both lack limbs but are able to move efficiently on land. Which mechanism BEST explains the similarity in their movement?
- A) Use of wings for balance
B) Use of fins for direction
C) Rhythmic contraction of body muscles
D) Use of hidden legs
10. A space satellite orbits the Earth at a constant speed and constant altitude. Why is it said that the Earth's gravity does 'zero work' on the satellite?
- A) Gravity is not a real force in space.
B) The satellite has no mass in space.
C) The satellite is moving too fast for work to be calculated.
D) The gravitational force acts perpendicular to the satellite's motion.
11. During a hot summer day, people feel uncomfortable even though the temperature is moderate. Which atmospheric factor is likely responsible?
- A) High oxygen content
B) High nitrogen content
C) High humidity content
D) Low carbon dioxide content

12. Sophie is trying to open a new jar of peanut butter. She grabs the lid and turns it repeatedly until it comes off the jar. Which simple machine is she utilizing to get the lid off?

- (A) Screw (B) Lever
(C) Pulley (D) Inclined plane

Section - 2

13. Reflex actions are sudden, automatic responses of the body to a stimulus. Which of the following are examples or characteristics of reflex actions?
- A) Withdrawing your hand immediately after touching a hot object.
B) Watering of the mouth at the sight of tasty food.
C) Solving a complex mathematical problem during an examination.
D) Blinking of the eyes when a bright light is suddenly flashed.
14. At a playground construction site, worker A uses a ramp to push a loaded wheelbarrow upward while worker B uses a rod and rock to lift a heavy boulder.
- A) Worker A is using an inclined plane to move things up easily.
B) Worker A is also using a wheel and axle (on the wheelbarrow) to make moving the heavy load smoother.
C) Worker B is using a lever, where the rock acts as the fixed point (fulcrum).
D) Worker B is using a wedge to slice the boulder in half.

15. Study the conversion process below:



Which of the following is true regarding processes P and Q?

- A) P is Freezing and Q is Condensation; both require cooling.
- B) P is Melting and Q is Evaporation; both require heating.
- C) P is Sublimation and Q is Boiling; both involve a decrease in molecular motion.
- D) P is Solidification and Q is Melting; both involve an increase in molecular motion.

16. The Ozone layer acts as a shield for life on Earth.

Which of the following statements about it are correct?

- A) It is found in the upper atmosphere and absorbs harmful Ultraviolet (UV) rays from the sun.
- B) UV rays can cause skin cancer, eye cataracts, and damage to small plants (phytoplankton).
- C) Chemicals called Chlorofluorocarbons (CFCs) are the primary cause of Ozone depletion.
- D) The Ozone layer is responsible for the blue color of the ocean.

17. A stone tied to a string is rotated in a circle at constant speed.

- A) Force is acting continuously
- B) Work done by the force is zero
- C) Energy is continuously required
- D) No force is acting

18. A sealed syringe contains air. When the piston is pushed, the volume decreases. Which statements are correct?

- A) Gas particles are far apart
- B) Gas has fixed volume
- C) Gas can be compressed
- D) Particles come closer on compression

Section - 3

COMPREHENSION-1

A village gets water from a nearby pond. The water is cloudy and contains small particles and some bacteria. The local health worker teaches the villagers how to make the water safe to drink.



19. Which method will kill bacteria and viruses in the pond water?

- A) Filtration
- B) Boiling
- C) Sedimentation
- D) Decantation

20. The villagers notice that sand and mud settle at the bottom if water is left undisturbed. This process is called:

- A) Filtration
- B) Decantation
- C) Sedimentation
- D) Chlorination

21. To make the water safe for long-term storage, which method is most suitable?

- A) Sedimentation only
- B) Filtration only
- C) Chlorination
- D) Using water in a water wheel

COMPREHENSION-2

Sai used a ramp to push a heavy box onto a platform, while Arjun lifted the same box directly upward. They observed that Sai applied less force but over a longer distance, while Arjun applied more force over a shorter distance.

22. What is the main advantage of using the ramp in this situation?

- A) It reduces both force and distance
- B) It eliminates gravitational force
- C) It reduces effort by increasing the distance over which force is applied
- D) It increases friction to help movement

23. Why did Arjun need to apply more force than Sai?

- A) Because lifting directly works against gravity over a shorter distance
- B) Because ramps increase friction
- C) Because gravitational force is absent on ramps
- D) Because muscular force is weaker on ramps

24. If the ramp surface becomes rough, what will most likely happen?

- A) Required force will decrease
- B) Friction will increase, making pushing harder
- C) Gravity will reduce
- D) Distance will decrease

COMPREHENSION-3

Ananya ate a meal containing rice, dal, and vegetables. As she started chewing, saliva mixed with the food. The food then moved to her stomach, where it was further broken down. Finally, nutrients were absorbed in

the small intestine and transported to different parts of the body.

25. Where does most digestion of food take place?

- A) Mouth
- B) Stomach
- C) Small intestine
- D) Large intestine

26. Why is chewing food properly important?

- A) It increases body temperature
- B) It helps in easy digestion
- C) It produces blood
- D) It cools the body

27. What is the role of saliva in the mouth?

- A) Absorbing nutrients
- B) Starting digestion of food
- C) Removing waste
- D) Producing energy

Section - 4

28.

Column A (Situation/Example)	Column B (Type/Importance)
A. A forest with dense trees, heavy rainfall, and high temperature	P. Maintains biodiversity
B. Animals like tigers, birds, and insects living together in forests	Q. Tropical rainforest
C. Planting trees in barren land	R. Deciduous Forest
D. Trees shedding leaves during winter in India	S. Afforestation

Which of the following options shows the correct matching?

- A) A – P, B – Q, C – R, D – S
- B) A – Q, B – P, C – S, D – R
- C) A – R, B – S, C – P, D – Q
- D) A – R, B – S, C – Q, D – P

29.

COLUMN A	COLUMN B
A) Malaria	P) Virus spread through air
B) Dengue	Q) Bacteria in contaminated food/water
C) Cholera	R) Mosquito (Anopheles) bite
D) Common Cold	S) Mosquito (Aedes) bite

- A) A-R, B-S, C-Q, D-P
- B) A-S, B-R, C-Q, D-P
- C) A-R, B-Q, C-S, D-P
- D) A-P, B-S, C-R, D-Q

30.

Column A	Column B
A Mercury	1. Spins in opposite direction & longest day
B Venus	2. Extreme temperature variation due to no atmosphere
C Jupiter	3. Bulges at equator due to rapid rotation
D. Uranus	4. Rotates on its side

- A) A-Q, B-P, C-R, D-S
- B) A-P, B-Q, C-R, D-S
- C) A-Q, B-R, C-P, D-S
- D) A-R, B-P, C-Q, D-S